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(54) INTEGRATION OF A PASSIVE COMPONENT IN A CAVITY OF AN INTEGRATED CIRCUIT PACKAGE

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(52) **U.S. Cl.**

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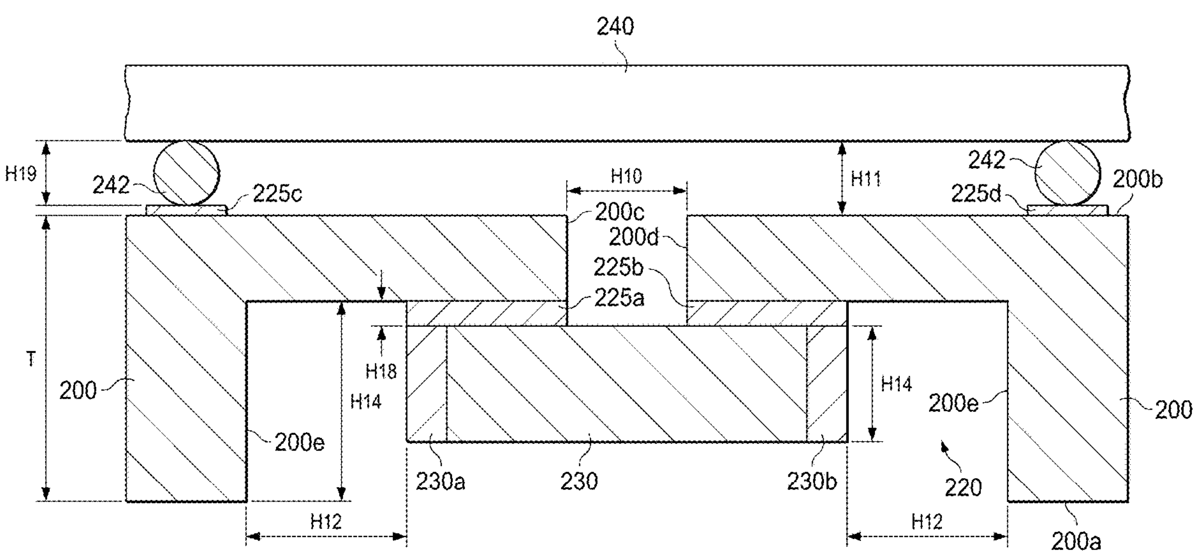
(57) **ABSTRACT**

A semiconductor device includes a lead frame, a semiconductor die, and an electronic component. The lead frame has opposing first and second surfaces. The semiconductor die is on the first surface of the lead frame. The electronic component is on the second surface of the lead frame.



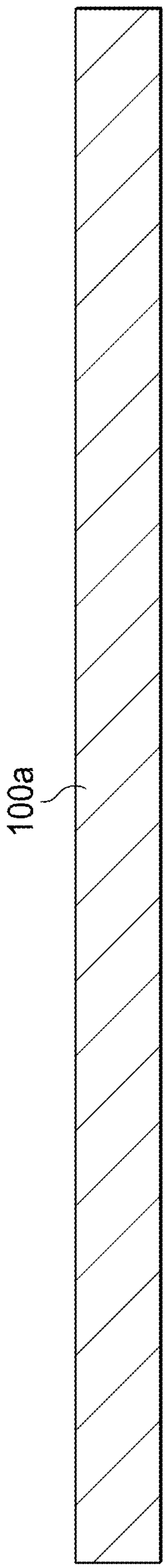


FIG. 1A

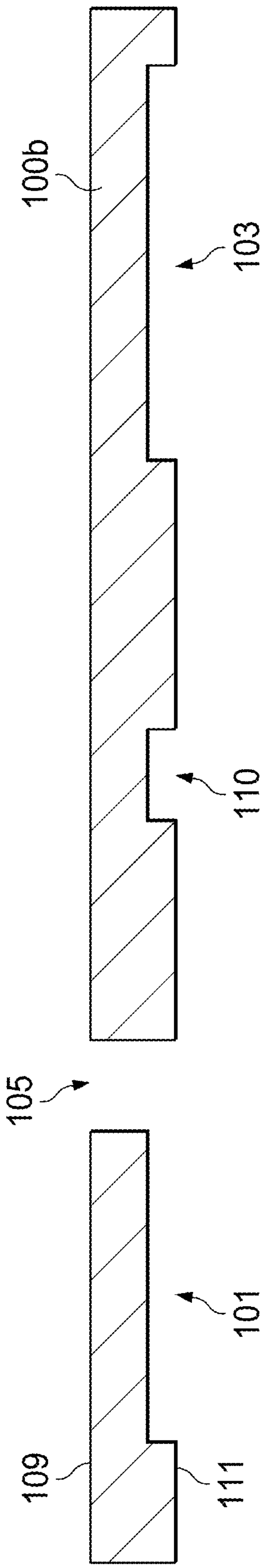


FIG. 1B

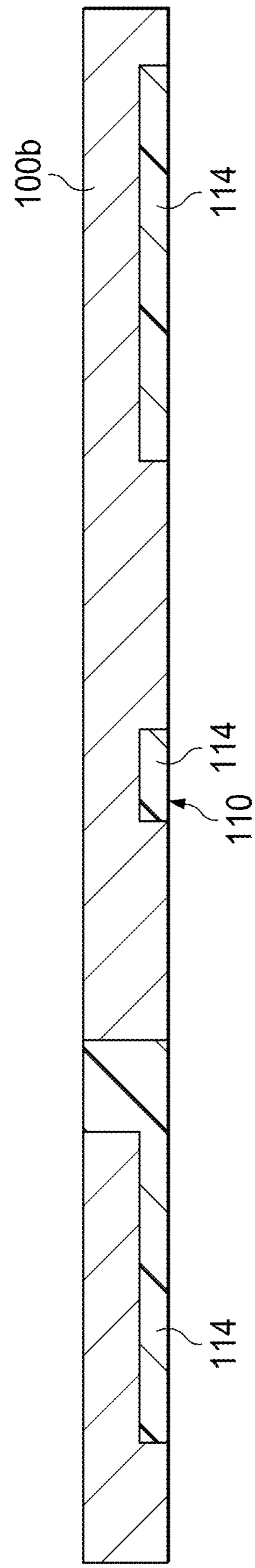


FIG. 1C

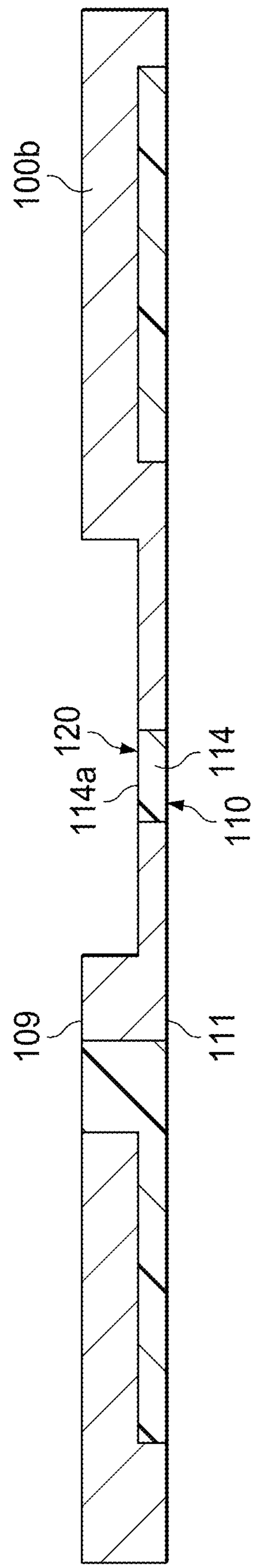

100b
109
111
120
114a
114
110

FIG. 1D

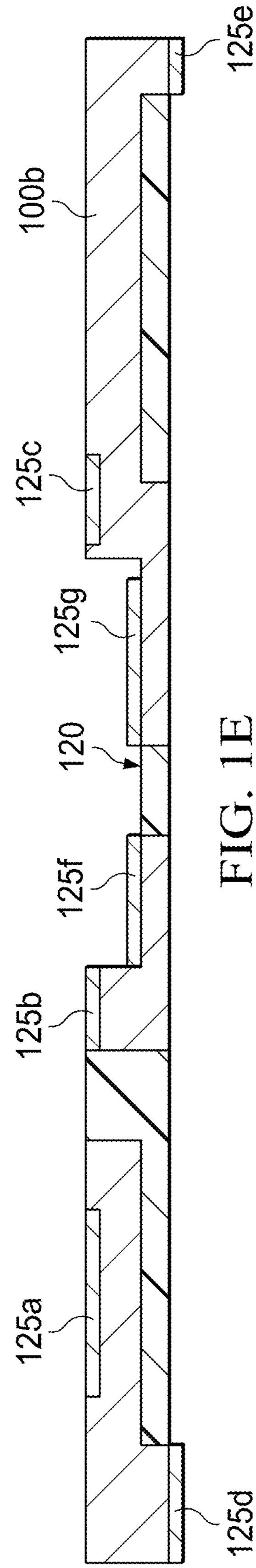

125e
100b
125c
125g
120
125f
125b
125a
125d

FIG. 1E

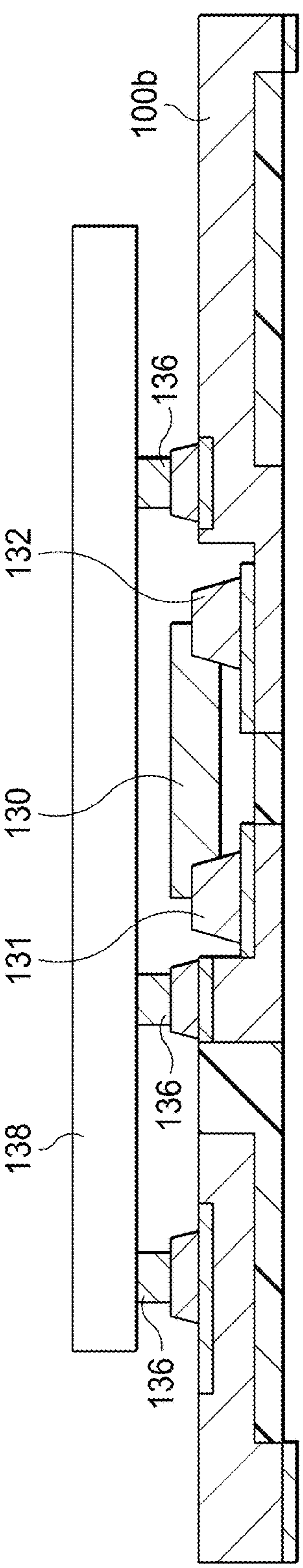

100b
136
132
130
131
136
138
136

FIG. 1F

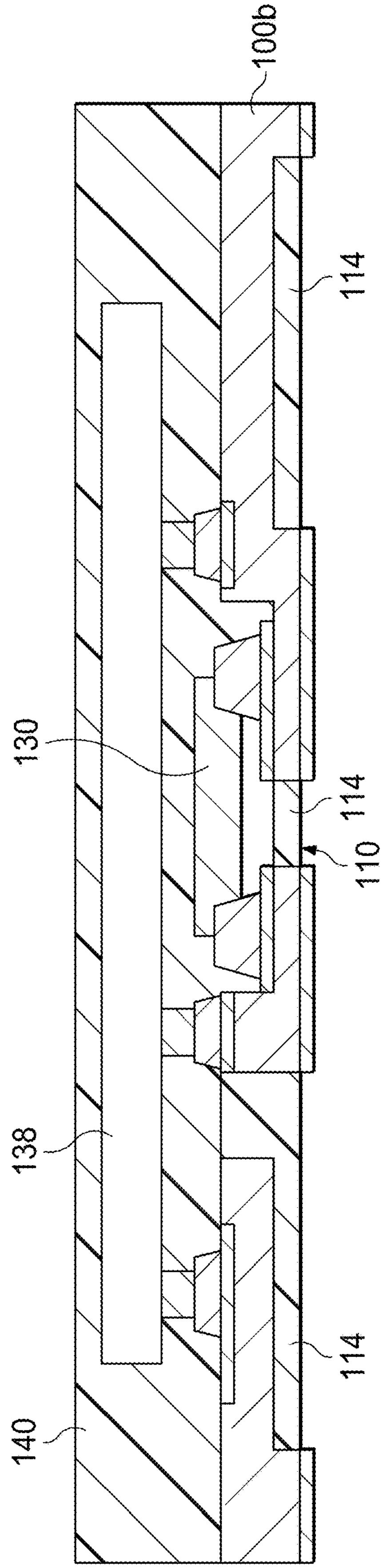

100b
114
130
114
110
138
114
140

FIG. 1G

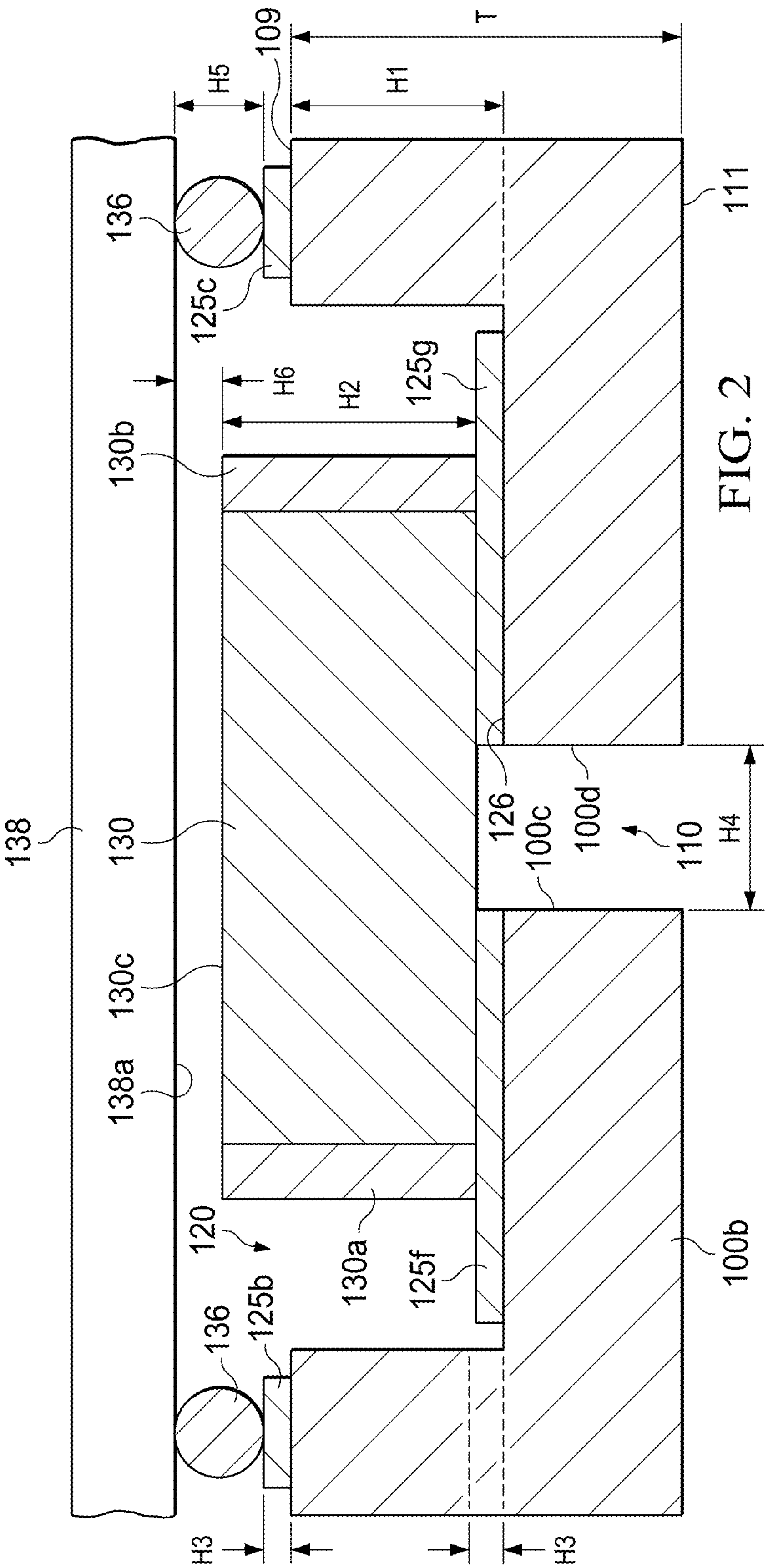

T
109
H5
H1
136
125c
111
125g
H6
H2
130b
138
130
126
100c
100d
110
H4
130c
138a
120
130a
125f
100b
136
125b
H3
H3

FIG. 2

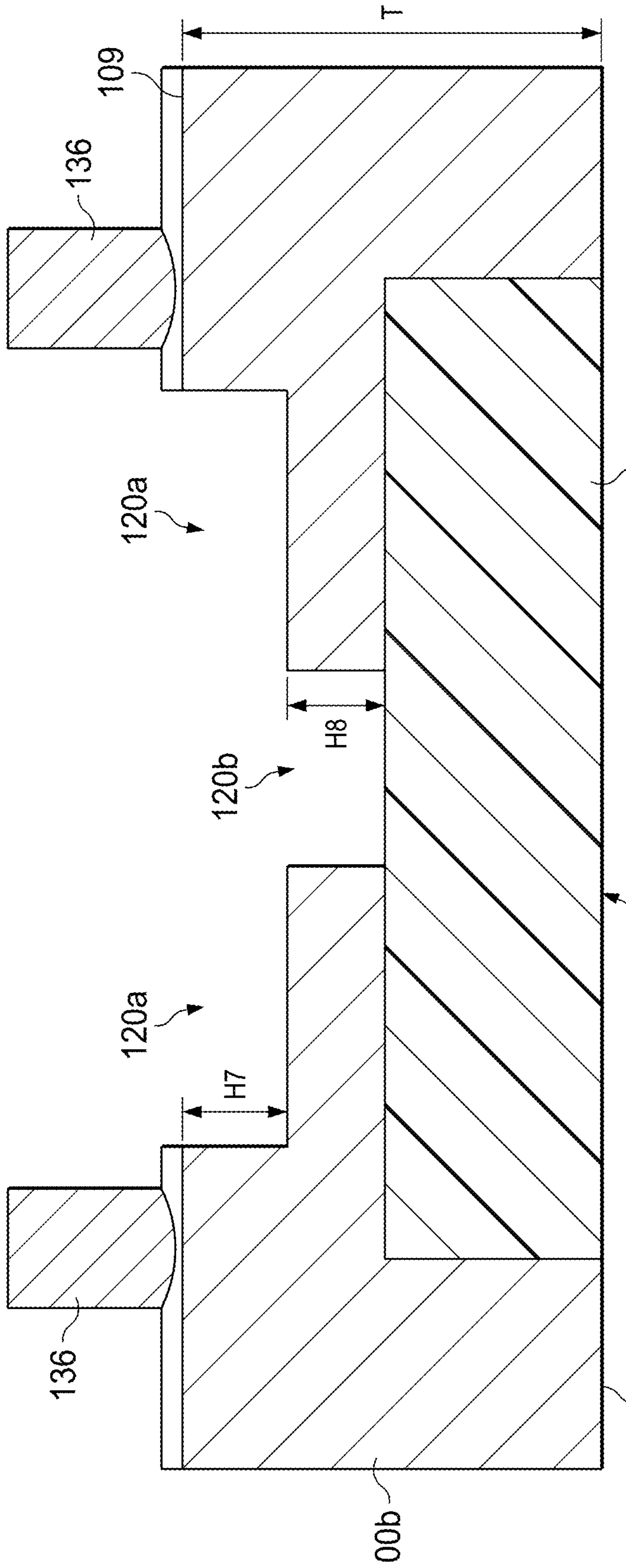


FIG. 3

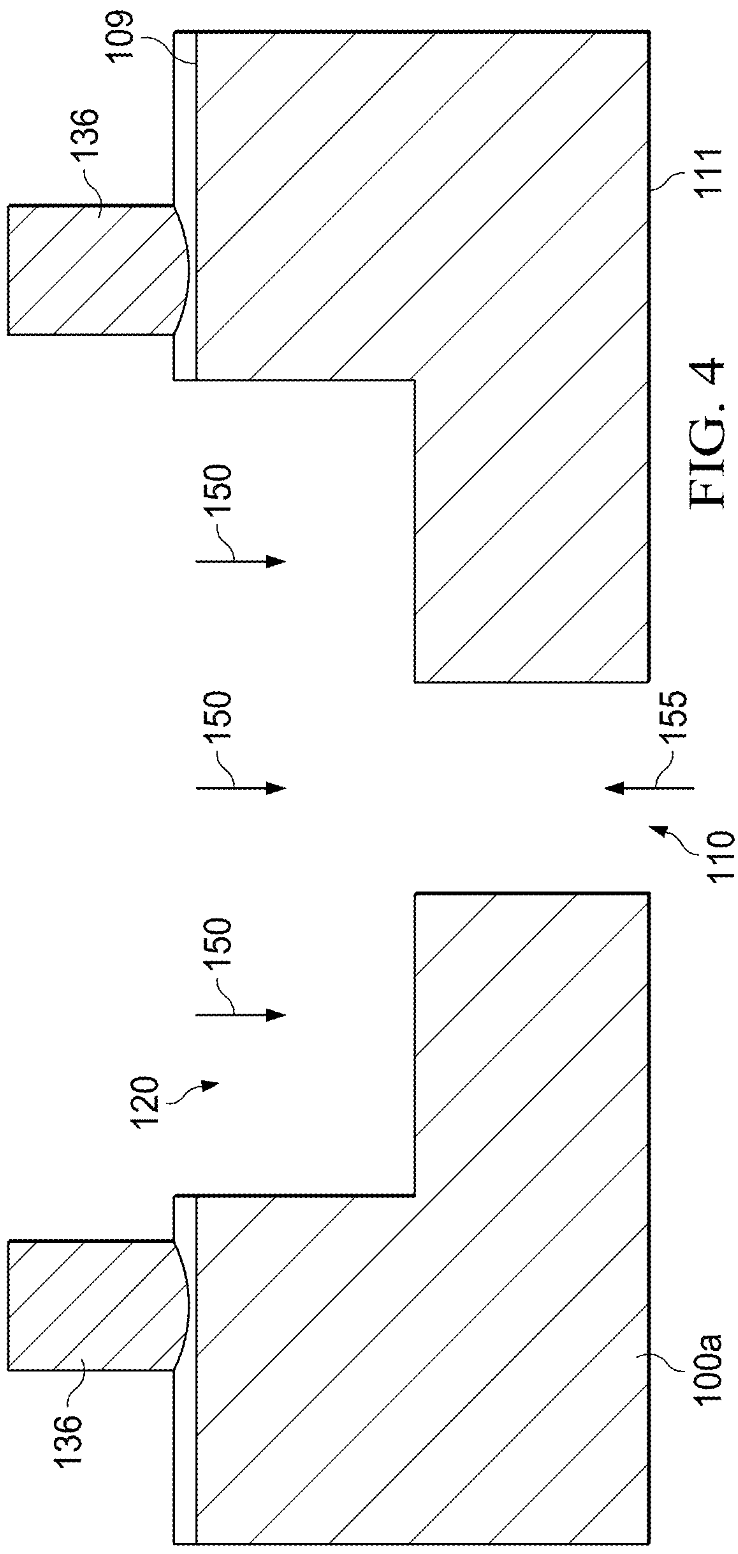


FIG. 4

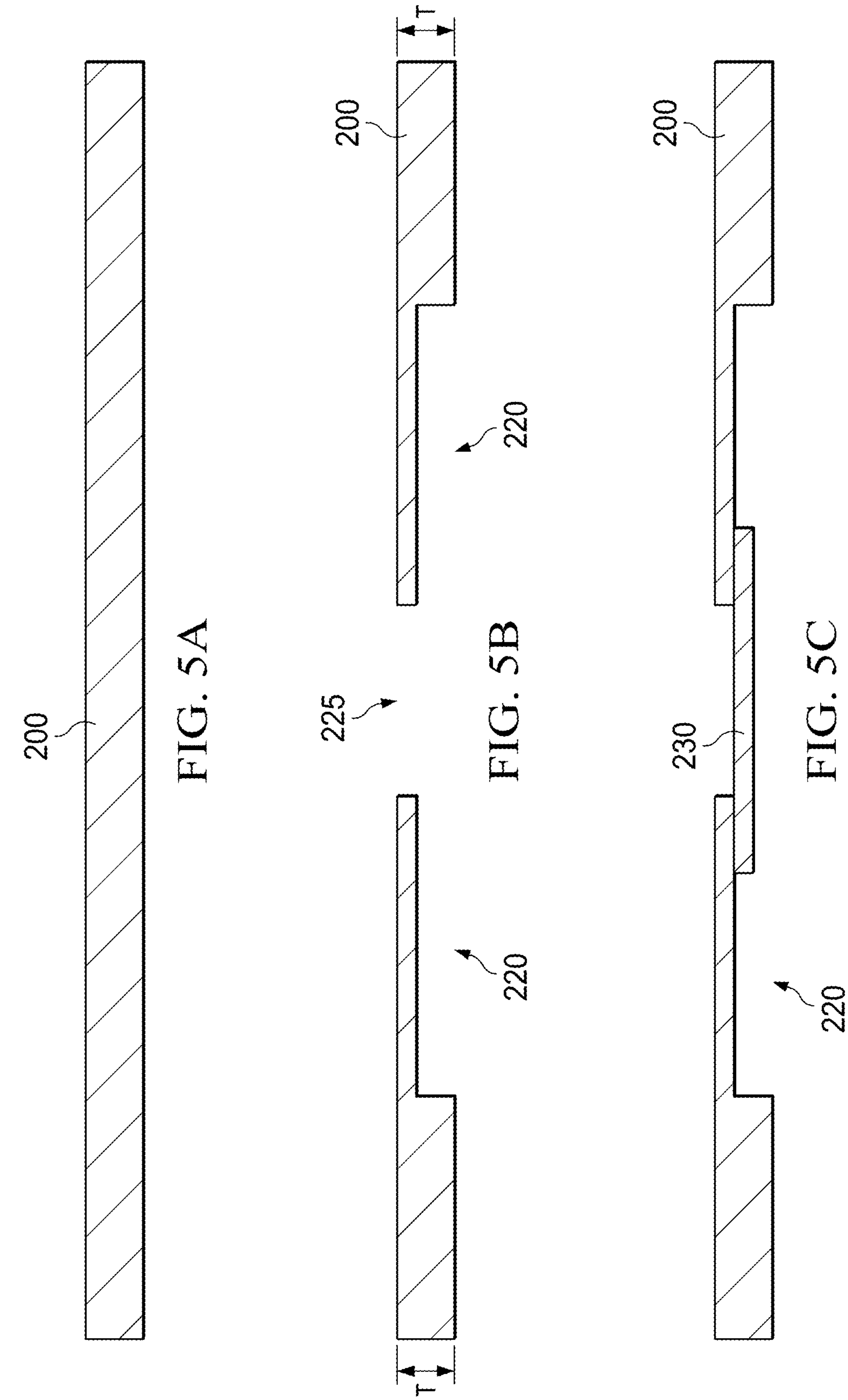


FIG. 5A

FIG. 5B

FIG. 5C

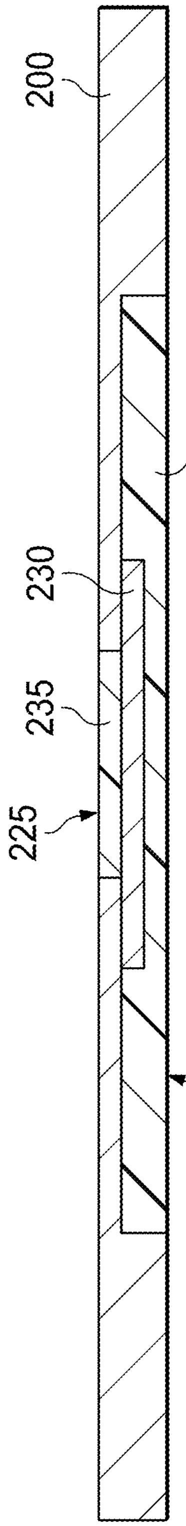


FIG. 5D

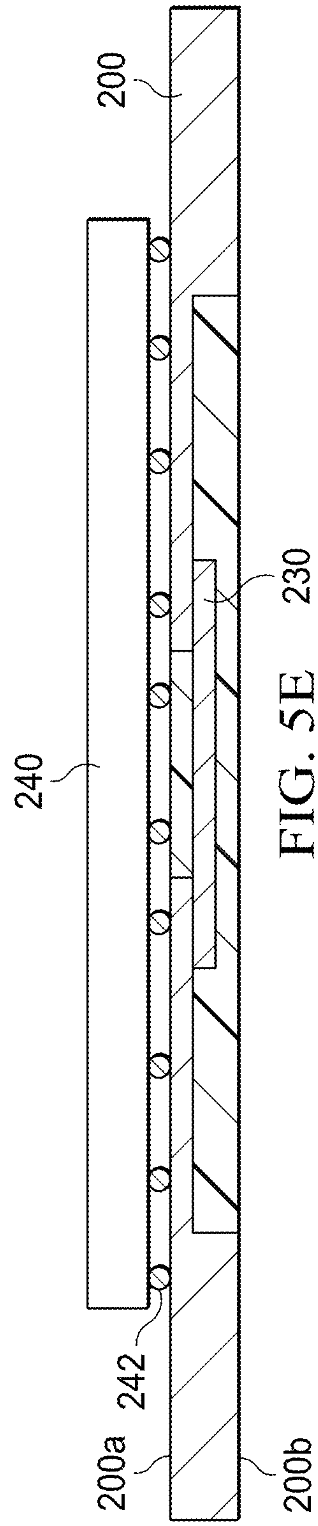
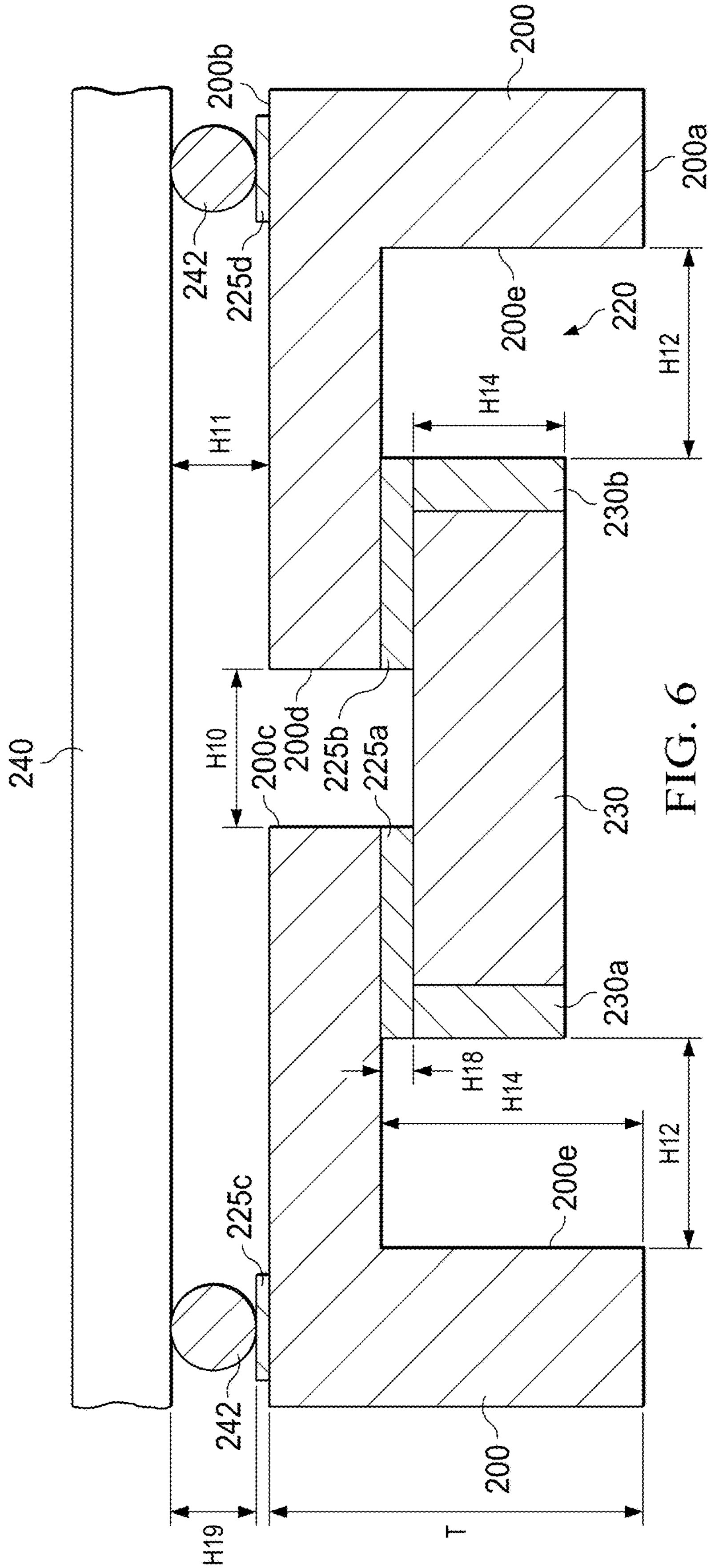


FIG. 5E



200
200b
200a
242
225d
200e
220
H12
H14
H11
230b
240
H10
200c
200d
225b
225a
230
230a
H18
H14
H12
200e
225c
242
200
H19
T

FIG. 6

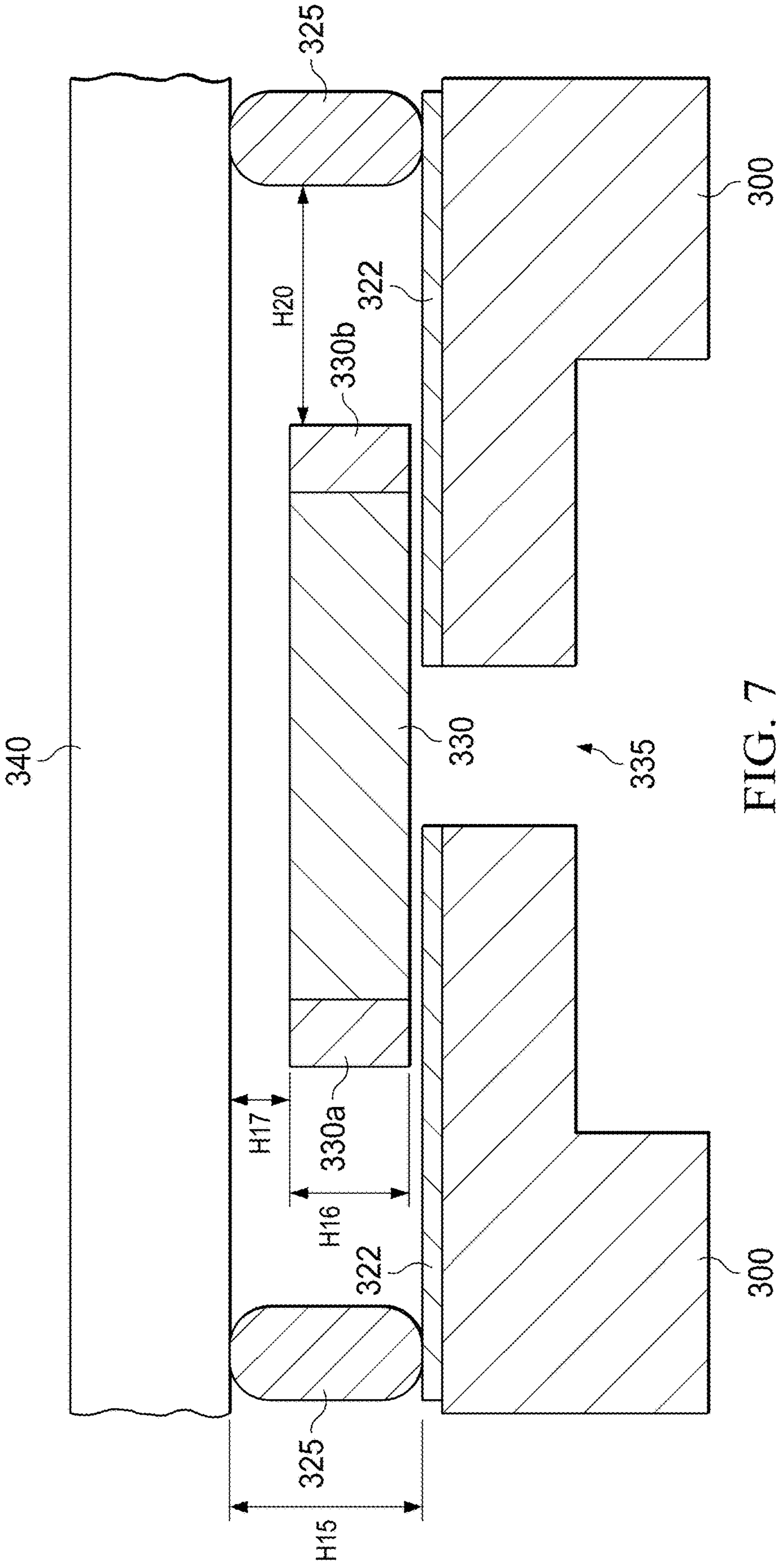


FIG. 7

INTEGRATION OF A PASSIVE COMPONENT IN A CAVITY OF AN INTEGRATED CIRCUIT PACKAGE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Division of U.S. patent application Ser. No. 15/950,984, filed Apr. 11, 2018, which claims priority to U.S. Provisional Patent App. No. 62/484,494 filed Apr. 12, 2017 and U.S. Provisional Patent App. No. 62/612,244 filed Dec. 29, 2017, which are incorporated herein by reference,

BACKGROUND

[0002] Many types of integrated circuits (ICs) have input/output (I/O) pins that are used to connect external passive or active components. An IC (also referred to as semiconductor die) often is attached to a lead frame and then surrounded by a mold compound to form a semiconductor package. The package is then attached to a printed circuit board (PCB). A capacitor (or other type of component) may be attached to the same PCB. Through traces on the PCB, the capacitor is electrically connected to one or more I/O pins of the lead frame (and through the lead frame to the IC). The connections between the capacitor and the components within the IC to which the capacitor is connected can create loop inductance which, in some applications such as power converters, can impact the performance of the IC.

[0003] In the case of a power converter, loop inductance may necessitate turning the power converter's power transistors on and off more slowly to reduce ringing. However, turning power transistors on and off more slowly results in greater switching losses.

SUMMARY

[0004] In described examples, a semiconductor package includes a lead frame, a semiconductor die attached to the lead frame, and a passive component (e.g., a capacitor) electrically connected to the semiconductor die through the lead frame. The lead frame includes a cavity in which at least a portion of the capacitor is disposed in a stacked arrangement. In one example, the cavity is formed in the lead frame on a side of the lead frame facing the semiconductor die.

[0005] Another example is directed to a method that includes etching a conductive member to form a lead frame for a semiconductor die and then etching the lead frame to form a cavity. The method further includes attaching a passive component to the lead frame inside the cavity and attaching the semiconductor die to the lead frame.

[0006] In yet another example, a semiconductor package includes a lead frame, a semiconductor die attached to the lead frame, and a capacitor electrically connected to the semiconductor die through the lead frame. The lead frame includes a cavity on a side of the lead frame facing the semiconductor die, and wherein at least a portion of the capacitor is disposed within the cavity of the lead frame.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIGS. **1**A-**1**G illustrate a process for forming a semiconductor package in which a passive component is disposed within a cavity in a lead frame in accordance with an embodiment.

[0008] FIG. **2** illustrates dimensions in accordance with the embodiment of FIGS. **1**A-**1**G.

[0009] FIG. **3** illustrates another embodiment of forming the cavity to include the passive component.

[0010] FIG. **4** illustrates yet another embodiment of forming the cavity to include the passive component.

[0011] FIGS. **5**A-**5**E illustrate a process for forming a semiconductor package in which a passive component is disposed within a cavity in the lead frame in accordance with an alternative embodiment.

[0012] FIG. **6** illustrates dimensions in accordance with the embodiment of FIGS. **5**A-**5**E.

[0013] FIG. **7** illustrates an embodiment of a semiconductor package in which a passive component is located between a semiconductor die and a lead frame without the use of a cavity.

DETAILED DESCRIPTION

[0014] The described embodiments pertain to a semiconductor package containing a semiconductor die attached to a lead frame an encapsulated in a mold. The package also includes a passive component. The package comprises a stacked configuration in which the passive component is attached to the lead frame either on the same side of the lead frame as the die or on the opposite side of the lead frame. In the embodiments in which the passive component is on the same side of the lead frame as the die, the passive component is mounted in a cavity formed in the lead frame, so the passive component is sandwiched between the die and the lead frame. In the embodiments in which the passive component is on the opposite of the lead frame from the die, the passive component is also mounted within a cavity formed in the lead frame. The passive component may have electrical contacts to the lead frame and, through the lead frame, connect to particular electrical contact pads to the semiconductor die.

[0015] By integrating the passive component into the package containing the semiconductor die in a stacked arrangement of the die, the passive component and the lead frame, the electrical conductors (between the passive component and the die) have reduced lengths, relative to lengths that would have existed if the passive component was outside the package altogether as described hereinabove. By reducing the length of the passive component-to-die conductors, the magnitude of the loop inductance advantageously is reduced.

[0016] In the embodiments described hereinbelow, the passive component is a capacitor. However, in other embodiments, the passive component can be other than a capacitor, such as a resistor or inductor. Further, an active component can be mounted in a stacked arrangement with the semiconductor die and lead frame, rather than a passive component.

[0017] FIGS. **1**A-**1**G illustrate an example of a process for fabricating an embodiment of a semiconductor package with a stacked configuration of a semiconductor die and capacitor. FIG. **1**A shows a cross-sectional view of a blank conductive member **100***a*. The conductive member may comprise copper alloy, or an alloy of another suitable conductive material. The blank conductive member **100** will be processed to form a lead frame **100***b* as described hereinbelow.

[0018] FIG. **1**B shows that the blank conductive member **100** has been partially etched to form a lead frame **100***b* comprising: an upper surface **109** containing portions to which the semiconductor die will be attached; and a lower surface **111** containing portions to mount to a PCB. In the example of FIG. **1**B, portions of the conductive material **100** have been removed to form recesses **101**, **103** and **105**. The recesses **101**, **103** and **105** are formed via any suitable process operation, such as etching the copper alloy. For example, copper can be etched by using a ferric chloride ($FeCl_3$)-based etching solution or cupric chloride ($CuCl_2$) in a complex base (e.g., pH greater than 7) solution. The $FeCl_3$ solution is generally more aggressive and faster than the $CuCl_2$ solution, but the $CuCl_2$ solution may be easier to control during manufacturing. A recess **110** also is formed as shown. This particular recess will function to separate portions of the lead frame that will be electrically connected to the capacitor to thereby avoid the capacitor's terminals from being shorted together.

[0019] FIG. **1**C illustrates the introduction of a pre-mold compound **114** into the recesses **101**, **103**, **105** and **110**. In some embodiments, the pre-mold compound **114** is a silica-based multi-functional aromatic resin. An example of the composition of the mold compound is approximately 60-80% silica, and the rest is a mixture of epoxy resins and additives, which may be included to modify specific properties.

[0020] FIG. **1**D illustrates that a cavity **120** has been etched in the lead frame **100***b*. A suitable etching, such as a copper etchant, can be used to etch the conductive material **100** to thereby form the cavity **120**. In one embodiment, the conductive material is etched from surface **109** (the surface that will face the die after installation) towards surface **101** (the surface of the lead frame opposite the die). The cross-sectional shape and size can be any suitable shape that accommodates the capacitor.

[0021] FIG. **1**E illustrates a selective metal plating process to thereby form conductive contacts **125***a*, **125***b*, **125***c*, **125***d*, **125***e*, **125***f* and **125***g* on various portions of the conductive material **100**. In this example, the conductive contacts **125***a*, **125***b* and **125***c* will connect the lead frame to the semiconductor die. The conductive contacts **125***d* and **125***e* will connect the lead frame to the PCB. Conductive contacts **125***f* and **125***g* are formed in the cavity **120** and will provide contact points for the capacitor.

[0022] In FIG. **1**F, the capacitor **130** has been placed in the cavity **120** and connected to the conductive contacts **125***f* and **125***g* via solder balls **131** and **132**. FIG. **1**F also shows the semiconductor die **138** mounted on the conductive material **100** (i.e., the lead frame) at conductive contacts **125***a*, **125***b* and **125***c* via copper (or other suitable conductive material) posts **136**. In this embodiment, at least a portion of the capacitor **130** is disposed within the cavity **120**. Due to the height created by the copper posts and conductive pads **125***a*-**125***c*, a portion of the capacitor **130** protrudes above the cavity as shown. In other embodiments, the entire capacitor is disposed within the cavity, with none of the capacitor protruding above the cavity.

[0023] FIG. **1**G illustrates the introduction of a post-mold compound **140** to encapsulate the semiconductor die **138** and capacitor **130** as shown to thereby form the semiconductor package. The post-mold compound **140** may be formulated from an epoxy resin containing inorganic fillers (e.g., fused silica), catalysts, flame retardants, stress modifiers, adhesion promoters, and other additives. In one example, a pelletized mold compound is liquefied and transferred to the recesses using a mold press. The liquefaction results in a low viscosity material that readily flows into the mold cavity and encapsulates the device.

[0024] FIG. **2** shows a portion of the semiconductor package of FIG. **1**G to identify various dimensions. Electrical contacts **130***a* and **130***b* bond to the conductive pads **125***f* and **125***g* respectively, as shown. The thickness of the lead frame between surfaces **109** and **111** is designated as T. The cavity **120** has been etched into the lead frame **100***b* from surface **109** to a depth designated as H**1**. The height of the capacitor is designated as H**2**. The height of the contact pads **125***b*, **125***c*, **125***f* and **125***g* is designated as H**3**. The spacing between lead frame portions **100***c* and **100***d* below the capacitor **130** is designated as H**4**, and the height of the copper posts **136** is designated as H**5**. In some embodiments, H**1** (the depth of the cavity **120**) is approximately 50% of T. In such embodiments, the lead frame is half-etched to form the cavity for the capacitor. The height H**2** of the capacitor is less than the combined heights H**1** (cavity depth), H**3** (height of conductive pads **125***b* and **125***c*) and H**5** (height of copper posts **136**). An additional clearance H**6** is also included between the top surface **130***c* of the capacitor **130** and the bottom surface **138***a* of the die **138** to allow for mold compound to flow. In one example, T is 200 micrometers ("microns"), H**1** (depth of cavity **120**) is 100 microns, H**3** is 15 microns, and H**5** is 60 microns. In this example, the distance from the cavity floor **126** to the bottom surface **138***a* of the die **138** is the combined heights of H**1**, H**3** and H**5**, or 175 microns. Thus, the height H**2** of the capacitor needs to be less than 175 microns in this example. For example, to permit a gap H**6** of 50 microns, the height H**2** of the capacitor **120** should be equal to or less than 125 microns. The length H**4** of the gap between lead frame portions **100***c* and **100***d* may be 170 microns.

[0025] As shown in FIG. **2**, an inherent gap exists between the top surface **130***c* of the capacitor **130** and the bottom surface **138***a* of the semiconductor die **138**, due to the height of the copper posts **136** and conductive pads **125***b* and **125***c*. Accordingly, in some embodiments, if the capacitor **130** is sufficiently thin, then the capacitor may be installed in that gap without the need for a cavity **120**. In some such cases, irrespective of whether a cavity is included, the gap can be increased by using taller copper posts **136** (e.g., 130 microns) to thereby permit the use of thicker capacitors.

[0026] In the example of FIG. **1**D, the recess **110** is formed by etching the conductive member **100***a* from surface **111** toward surface **109**, and etching 50% of the thickness of the conductive member **100***a*. In FIG. **1**D, the cavity **120** is then formed by etching the lead frame **100***b* from the surface **109** in the direction of opposing surface **111**, and etching 50% of the thickness of the lead frame. The etching is performed until the etching tool just reaches the top surface **114***a* of the premold compound **114**.

[0027] FIG. **3** shows an alternative embodiment for forming the cavity **120**. In this example, the etching tool has etched the lead frame **100***b* from surface **109** in the direction of surface **111**, but less than 50% of the thickness T of the lead frame. Two etching processes are performed in this embodiment. A first etching process is performed to a depth of H**7** to form a cavity **120***a*. A second etching process is then performed in a central region of cavity **120***a* to a depth of H**8** to thereby electrically isolate the lead frame contacts for the capacitor. In one embodiment, H**7** is between approximately 20% and 40% of T, and H**8** is T minus H**7**. If the recess **110** containing the mold compound is formed to a depth of 50% of T, and H7 is 20%-40% of T, then H**8** is 10%-30% of T.

[0028] In the example of FIGS. **1**B through **1**D, recess **110** is formed and then filled with premold compound **114**, and then cavity **120** is etched down to the level of the mold compound. FIG. **4** illustrates an alternative embodiment in which the recess **110** and cavity **120** are formed before the mold compound is introduced into the recess **110**. In this embodiment, a two direction etching machine is used to etch the conductive member **100***a* from both surfaces **109** and **111** in the direction of the other surface. Accordingly, cavity **120** is formed by etching the conductive member **100***a* in the direction of arrows **150**, while recess **110** is formed by etching the conductive member **100***a* in the direction of arrow **155**. The etching process is completed after the recess **110** and cavity **120** are fully formed. This process for forming the cavity **120** is sufficient, even if the etching is less accurate than another embodiment in which recess **110** was already formed and filled with the premold compound **114**.

[0029] FIGS. **5**A-**5**E illustrate an alternative embodiment for forming the semiconductor package. In this embodiment, the cavity is on a side of the lead frame opposite the semiconductor die, so the cavity is not sandwiched between the lead frame and the die. The process starts with a blank conductive member **200** at FIG. **5**A. After the conductive member **200** has been etched and processed, it functions as the lead frame for the semiconductor die. The conductive member may comprise a copper alloy, or other suitable material. FIG. **5**B illustrates the formation of a cavity **220** in the conductive member **200**, in which the capacitor will be inserted. In this embodiment, the cavity **220** may be etched to a depth of approximately 75% of the thickness T of the conductive member **200**. Also, a recess **225** is formed to provide the electrical isolation between the capacitor's contacts.

[0030] FIG. **5**C illustrates that the capacitor **230** has been attached to the conductive member **200** inside the cavity. In this embodiment, the height of the capacitor **230** is equal to or less than the height of the cavity. Accordingly, no portion of the capacitor **230** protrudes out of the cavity. FIG. **5**D illustrates that a premold compound **235** is introduced into the cavity **220** and recess **225** and other recesses and cavities formed in the conductive member **200** to form the lead frame. FIG. **5**E shows a semiconductor die **240** attached via copper posts or solder balls to a surface **200***a* of the lead frame, opposite a surface **200***b* in which the cavity **220** was etched to accommodate the capacitor **230**. Although not shown, the semiconductor die **240** and lead frame are encapsulated in a post-mold compound to form the finished semiconductor package.

[0031] FIG. **6** shows a portion of the semiconductor package of FIG. **5**E to identify various dimensions. Electrical contacts **230***a* and **230***b* of the capacitor **230** bond to the conductive pads **225***a* and **225***b* respectively, as shown. The thickness of the lead frame is designated as T as described hereinabove (although the lead frame thickness need not be the same in every embodiment). The cavity **220** has been etched into the lead frame **200** from surface **200***a* toward surface **200***b* to a depth designated as H**14**. The height of the capacitor is designated as H**13**. The height of the contact pads ***225a, 225b,*** **225***c* and **225***d* is designated as H**18**. The spacing of the recess between lead frame portions **200***c* and **200***d* above the capacitor **230** is designated as H**10**, and the height of the copper posts **242** is designated as H**19**. In some embodiments, H**14** (the depth of the cavity **220**) is approximately 75% of T. In such embodiments, the lead frame is three-quarter-etched to form the cavity **220** for the capacitor **230**. The height H**13** of the capacitor is less than the height H1 of cavity in some embodiments, so that no portion of the capacitor protrudes out of the cavity **220**. In other embodiments, a portion of the capacitor **230** does protrude out of the cavity **230**. An additional clearance H**11** (e.g., **75** microns) is also included between the top surface **200***b* of the lead frame **200** and the bottom surface of the die **240** to allow for mold compound to flow. In one example, T is 200 micrometers, H**14** (depth of cavity **120**) is 150 microns, H**18** is 15 microns, and H**19** is 60 microns. In this example, the height H**13** of the capacitor should be 150 microns (e.g., less than or equal to 135 microns). The length H**10** of the gap between lead frame portions **200***c* and **200***d* may be 125 microns. The distance H**12** between the ends of the capacitor **230** and the side walls **200***e* of the cavity **220** may be approximately 150 microns.

[0032] As shown in FIG. **2**, an inherent gap exists between the top surface **130***c* of the capacitor **130** and the bottom surface **138***a* of the semiconductor die **138**, due to the height of the copper posts **136** and conductive pads **125***b* and **125***c*. Accordingly, in some embodiments, if the capacitor **130** is sufficiently thin, then the capacitor may be installed in that gap without the need for a cavity **120**. In some such cases, irrespective of whether a cavity is included, the gap can be increased by using taller copper posts **136** (e.g., 130 microns) to thereby permit the use of thicker capacitors.

[0033] FIG. **7** shows an example of a semiconductor package in which a passive component **330** is located between a semiconductor die **340** and a lead frame **300**. The passive component may be a capacitor or other type of passive electrical device (e.g., an inductor). In this example, the passive component **330** is not placed within a cavity formed in the lead frame **300**. Without the need to etch a cavity for the purpose of placing the passive component, the etching process (of embodiments described hereinabove) is avoided. Instead, solder posts **325** are provided with a height H**15** that is sufficiently large to accommodate the height of the passive component. In one example, H**15** is approximately 130 microns, in which case the height H**16** of the passive component is less than 130 microns (e.g., approximately 80 microns). The dimension H**20** represents the spacing between the passive component and the nearest solder post **325**. In one example, H**20** is approximately 150 microns.

[0034] Opposing ends of the passive component **330** include electrical contacts **330***a* and **330***b* Electrical contacts **330***a* and **330***b* bond to conductive pads **322** as shown to thereby electrical connect the passive component to the lead frame, and through solder posts **325** to the semiconductor die **340**. A recess **335** is formed in the lead frame **300** to electrically isolate the electrical contacts **330***a* and **330***b* of the passive component **330**.

[0035] In the example of FIG. **7**, a space with a dimension H**17** is provided between the passive component **330** and the semiconductor die **340**. The space between the passive component and the semiconductor die permits a post-mold compound to encapsulate the semiconductor die **340** and passive component **330** to thereby form the semiconductor package. As described hereinabove, the post-mold compound may be formulated from an epoxy resin containing inorganic fillers (e.g., fused silica), catalysts, flame retardants, stress modifiers, adhesion promoters, and other additives. In one example, a pelletized mold compound is liquefied and transferred to the recesses using a mold press device.

[0036] In example embodiments, the term "approximately" means that a value or range of values is either a stated value or range of values or within plus or minus 10% from that stated value or range of values.

[0037] In this description, the term "couple" or "couples" means either an indirect or direct wired or wireless connection. Thus, if a first device couples to a second device, that connection may be through a direct connection or through an indirect connection via other devices and connections. Also, in this description, the recitation "based on" means "based at least in part on." Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

[0038] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

What is claimed is:

1. A semiconductor device, comprising:

a lead frame having opposing first and second surfaces;

a semiconductor die on the first surface of the lead frame; and

an electronic component on the second surface of the lead frame.

2. The semiconductor device of claim **1**, wherein the lead frame has a cavity, the second surface is part of the cavity, and the electronic component is in the cavity.

3. The semiconductor device of claim **2**, wherein the lead frame has an opening that extends between the first and second surfaces and connects with the cavity.

4. The semiconductor device of claim **3**, wherein the electronic component has a first terminal and a second terminal, the first terminal electrically coupled to a first portion of the lead frame on a first side of the opening, the second terminal electrically coupled to a second portion of the lead frame on a second side of the opening, the first side opposing the second side.

5. The semiconductor device of claim **3**, further comprising a pre-mold compound that fills at least a part of the opening.

6. The semiconductor device of claim **5**, further comprising a post-mold compound covering at least parts of the semiconductor die and the lead frame.

7. The semiconductor device of claim **6**, where the post-mold compound and the pre-mold compound have different materials.

8. The semiconductor device of claim **1**, wherein a portion of the electronic component sits within the cavity, and a remainder of the electronic component is outside the cavity and between the lead frame and the semiconductor die.

9. The semiconductor device of claim **1**, wherein the lead frame has a thickness T, and the cavity has a depth of approximately 50% of T.

10. The semiconductor device of claim **1**, wherein the lead frame has a thickness T, and the cavity has a height that is between approximately 20% and 40% of T.

11. The semiconductor device of claim **1**, wherein no portion of the electronic device protrudes out of the cavity.

12. The semiconductor device of claim **1**, wherein the electronic component includes at least one of: a capacitor, an inductor, or a resistor.

13. The semiconductor device of claim **1**, further comprising metal interconnects coupled between the semiconductor die to the lead frame.

14. The semiconductor device of claim **13**, wherein the metal interconnects include copper posts.

15. The semiconductor device of claim **1**, further comprising metal pads on the second surface, and the semiconductor die is mounted on the metal pads.

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